

FINA2204 Tutorial 6: Properties of Stock Options

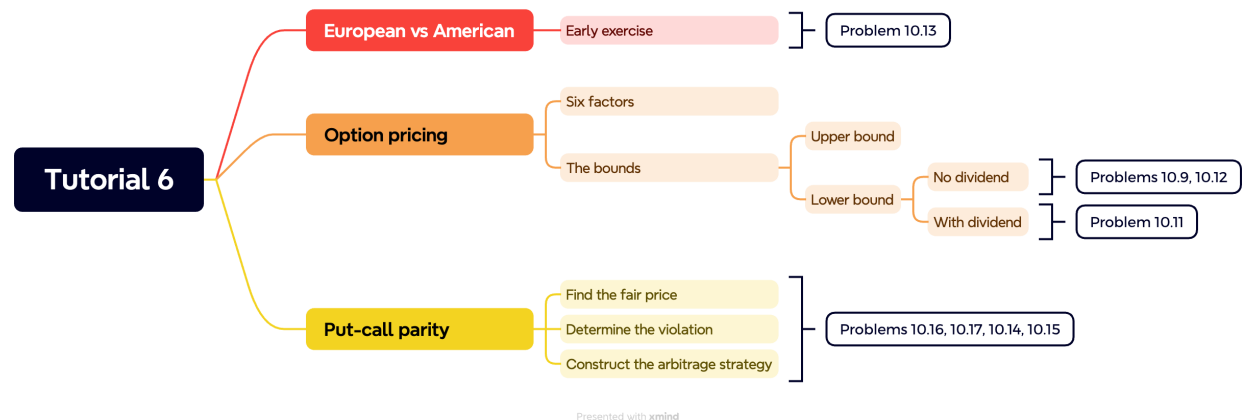
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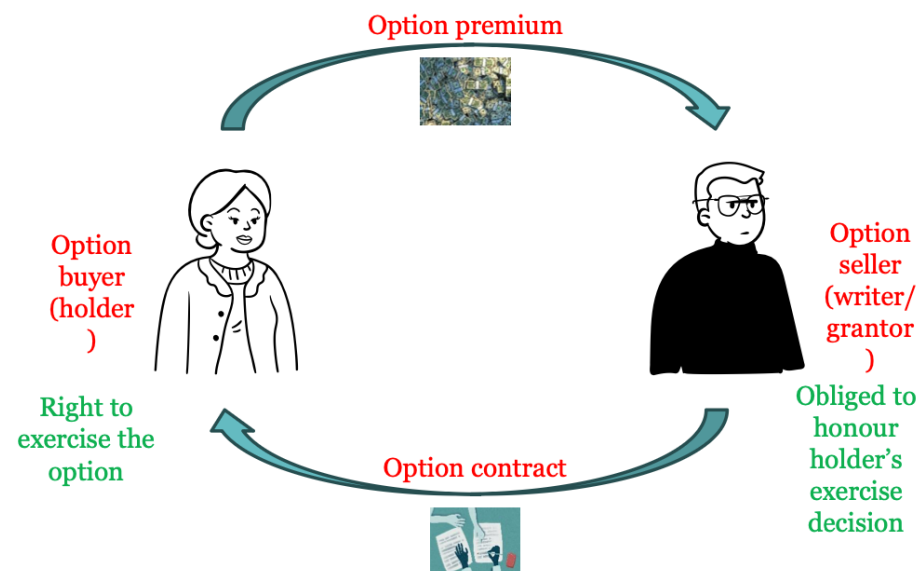
Agenda

- Problem 10.9
- Problem 10.11
- Problem 10.12
- Problem 10.13
- Problem 10.14
- Problem 10.15
- Problem 10.16
- Problem 10.17



Recap: Options

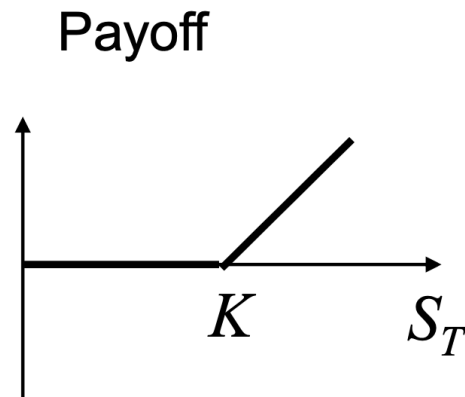
- An **option** is a contract that gives the holder **the right, but not the obligation**, to buy or to sell an underlying asset at a predetermined price on or before a specified date.



Recap: Option payoff diagrams

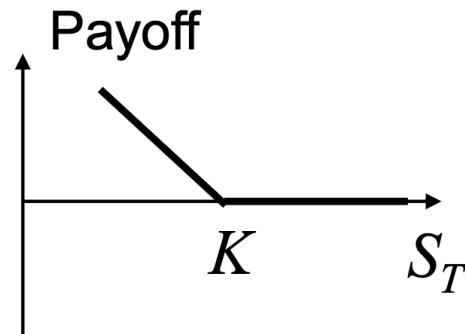
Long Call

$$\text{Payoff} = \max(S_T - K, 0)$$

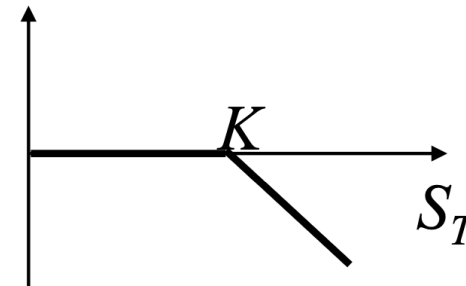


Long put

$$\text{Payoff} = \max(K - S_T, 0)$$



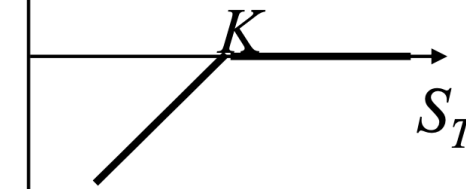
Payoff



Short call

$$\text{Payoff} = -\max(S_T - K, 0) = \min(K - S_T, 0)$$

Payoff



Short put

$$\text{Payoff} = -\max(K - S_T, 0) = \min(S_T - K, 0)$$

Recap: Factors affecting option price

• Notation

- c : European call option price
- p : European put option price
- S_0 : Stock price today
- K : Strike price
- T : Life of option
- σ : Volatility of stock price
- C : American Call option price
- P : American Put option price
- S_T : Stock price at option maturity
- D : Present value of dividends during option's life
- r : Risk-free rate for maturity T with cont. comp.

Recap: European vs. American

1. European option (vanilla, standard): An option can **only** be exercised on the expiration date.
2. American option: An option can be exercised on **any date** before the expiration date (*early exercise*).
 - The American option gives you all the rights of the European option plus **extra flexibility** (you can exercise earlier).
 - **Takeaway:** *An American option is **always worth at least as much as** a European option on the same asset with the same strike price and exercise date.*

Recap: Early exercise

- Early exercise always gives up the option's remaining time value (TV).
- **Higher volatility** → **higher TV** : more uncertainty makes the option's flexibility more valuable.
- **Lower volatility** → **lower TV** : less value is lost by exercising early, so early exercise becomes **more attractive**.
- Call: exercising early means paying K earlier → lose interest rK .
- **Higher risk-free rate** → **larger interest loss** → early exercise becomes less attractive.

Recap: Early exercise

- Put: exercising early means receiving K earlier $\rightarrow$ gain interest rK .
- Higher risk-free rate $\rightarrow$ larger interest gain $\rightarrow$ early exercise becomes **more attractive**.
- Call: exercising early before the ex-dividend date $\rightarrow$ receive dividend D .
- Higher dividend $\rightarrow$ larger dividend gain $\rightarrow$ early exercise becomes **more attractive**.

Recap: Early exercise

- Put: waiting through the ex-dividend date → benefit from the dividend-induced stock price drop.
- Higher dividend → larger benefit from waiting → early exercise becomes less attractive.

Recap: Early exercise

- Should American options be exercised early or not?

American options	Non-dividend-paying	Dividend-paying
Call	Never (because $0 > rK + TV$ can never hold)	Yes, if $D > rK + TV$
Put	Yes, if $rK > TV$	Yes, if $rK > D + TV$

Problem 10.13

- *Give an intuitive explanation of why the early exercise of an American put becomes more attractive as the risk-free rate increases and volatility decreases.*

Answer 10.13

- The early exercise of an American put is attractive when the interest earned on the strike price is greater than the insurance element lost ($rK > TV$).
- When interest rates increase, the value of the interest earned on the strike price increases, making early exercise more attractive.
- When volatility decreases, the insurance element is less valuable. Again, this makes early exercise more attractive.

Recap: Upper and lower bounds

	European call c	European put p
Upper bound (both)	?	?
Lower bound. no dividend	?	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	?	?
Lower bound. no dividend	?	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Recap: Upper and lower bounds

- **Upper bounds for calls**

An American or European call option gives the holder the right to buy one share of a stock for a certain price. No matter what happens, the option can never be worth more than the stock. Hence, the stock price is an upper bound to the option price:

$$c \leq S_0 \quad \text{and} \quad C \leq S_0 \quad (10.1)$$

If these relationships were not true, an arbitrageur could easily make a riskless profit by buying the stock and selling the call option.

Source: textbook, p.250

Recap: Upper and lower bounds



Powerball Draw 1583

4d 2h 38m 12s
until the chance to
win
\$12 million

[Play now](#)

[Join now](#)



Would anyone pay **more than \$12 million** for one ticket?

Recap: Upper and lower bounds

- **Arbitrage:** A strategy that yields a **positive** cash flow at one point in time and **non-negative** cash flows at all other points in time.
- In option pricing, arbitrage can occur if the option price goes above or below a certain range (*the upper and lower bounds*), traders can construct arbitrage strategies.

Strategy	CF_0	CF_1	CF_2
A	+2	+1	-1
B	0	+1	0
C	-1	+2	+4
D	-1	+5	0
E	+1	0	0

Recap: Upper and lower bounds

Case 1: European / American call option upper bound

- Suppose the stock price (S_0) is \$100, Strike price: K (irrelevant).
- If a call option is priced at $C = 120$ (this **violates** the condition $C \leq S_0$)

Arbitrageur's actions

1. Sell the call option (arbitrageur sells one call option in the market; receives $C = 120$ in cash from the buyer).
2. Meanwhile, arbitrageur buys the stock in the market for $S_0 = 100$.

Recap: Upper and lower bounds

- In this case, arbitrageur now owns one share of stock and has extra cash left over ($120 - 100 = 20 > 0$). And stock must be delivered if the option buyer exercises.

Two possible scenarios: If $S_T \geq K$, the option buyer will pay K and receive the stock. If $S_T < K$, the option buyer does nothing.

Strategy	t=0	@T $S_T \geq K$	@T $S_T < K$
Sell call	+120	$-(S_T - K)$	0
Buy stock	-100	$+S_T$	$+S_T$
Net CF	+20	K	S_T

Positive

Non-negative

Recap: Upper and lower bounds

- In both cases, the arbitrageur has guaranteed the extra 20 profit from the mispricing.

Why this forces $C \leq S_0$?

- Everyone would repeat this strategy (sell the call, buy the stock).
- The selling pressure on calls would push call price down.
- The buying pressure on stocks would push the stock price up.
- Market forces restore the condition $C \leq S_0$.

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	?
Lower bound. no dividend	?	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	?
Lower bound. no dividend	?	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Recap: Upper and lower bounds

- **Upper bounds for puts**

Q&A: Why can we discount the strike for a European put, but not for an American put?

An American put option gives the holder the right to sell one share of a stock for K . No matter how low the stock price becomes, the option can never be worth more than K . Hence,

$$P \leq K \quad (10.2)$$

For European options, we know that at maturity the option cannot be worth more than K . It follows that it cannot be worth more than the present value of K today:

$$p \leq Ke^{-rT} \quad (10.3)$$

If this were not true, an arbitrageur could make a riskless profit by writing the option and investing the proceeds of the sale at the risk-free interest rate.

Source: textbook, p.250

Recap: Upper and lower bounds

European put (upper bound = Ke^{-rT})

- Can only be exercised at maturity.
- Seller's obligation to pay arises only at that time.
- Seller can *set aside the present value of K today*, invest risk-free, and cover the payment later.

American put (upper bound = K)

- Can be exercised **any time** before expiration. Since the obligation may arise at any time, the seller must *hold the full strike amount in cash* to ensure readiness for the counterparty's exercise.

Recap: Upper and lower bounds

- **Case 2.1: European put option upper bound**
- Suppose the strike price (K) is \$100. If the market price (p) of put option is \$120.
- Time to maturity $T = 1$ year, risk-free rate $r = 5\%$. So present value of strike $= Ke^{-rT} = 100e^{-0.05 \times 1} = 95.12$
- Market price of put $p = 120 > 95.12$

Arbitrageur's actions

1. Write the put option, receive 120 in cash, with the obligation to buy at strike if exercised.
2. Immediately invest the present value of strike 95.12 at risk-free rate for one year (e.g., zero-coupon bond).

Recap: Upper and lower bounds

- Two possible scenarios at maturity
 1. If $S_T < K$: put buyer exercises; arbitrageur buys stock at S_T and pays K ; If $S_T \geq K$: Put expires worthless. Arbitrageur keeps the full 120.
- Market forces push option price down so $P \leq Ke^{-rT}$.

Strategy	t=0	@T $S_T \geq 100$	@T $S_T < 100$
Sell put	+120	0	$-(100 - S_T)$
Lend Ke^{-rT}	-95.12	+100	+100
Net CF	+24.88	+100	S_T

Positive

Non-negative

Recap: Upper and lower bounds

Case 2.2: American put option upper bond

- Suppose the strike price (K) is \$100. If the market price (P) of put option is \$120 (this **violates** the condition $P \leq K$).

Arbitrageur's actions

- Write the put option, receive 120.
- But since exercise can occur anytime, it is equivalent to setting aside 100 for the strike payment, leaving a net flow of 20.

Recap: Upper and lower bounds

- Two possible scenarios: If $S_T < 100$: put buyer **exercises**; arbitrageur buys the stock at S_T and pays K ; If $S_T \geq 100$: put expires worthless.

Strategy	t=0	@T $S_T \geq 100$	@T $S_T < 100$
Sell put	+120	0	$-(100 - S_T)$
Set aside cash	-100	+100	+100
Net CF	+20	+100	$+S_T$

- Market forces push option price down so $P \leq K$.

Positive

Non-negative

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	?	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	?	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Lower bounds for call on non-dividend-paying stocks

Portfolio A: one European call option plus a zero-coupon bond that provides a payoff of K at time T $\rightarrow V_A(0) = c + Ke^{-rT}$

Portfolio B: one share of the stock $\rightarrow V_B(0) = S_0$

In portfolio A, the zero-coupon bond will be worth K at time T . If $S_T > K$, the call option is exercised at maturity and portfolio A is worth S_T . If $S_T < K$, the call option expires worthless and the portfolio is worth K . Hence, at time T , portfolio A is worth

$$[\max(S_T - K, 0) + K] \rightarrow \max(S_T, K)$$

Portfolio B is worth S_T at time T . Hence, portfolio A is always worth as much as, and can be worth more than, portfolio B at the option's maturity. It follows that in the absence of arbitrage opportunities this must also be true today. The zero-coupon bond is worth Ke^{-rT} today. Hence,

$$c + Ke^{-rT} \geq S_0$$

or

$$c \geq S_0 - Ke^{-rT}$$

Because the worst that can happen to a call option is that it expires worthless, its value cannot be negative. This means that $c \geq 0$, so that

$$c \geq \max(S_0 - Ke^{-rT}, 0) \quad (10.4)$$

If A always gives at least as much terminal wealth as B, A cannot cost less today!

Source:
textbook, p.251

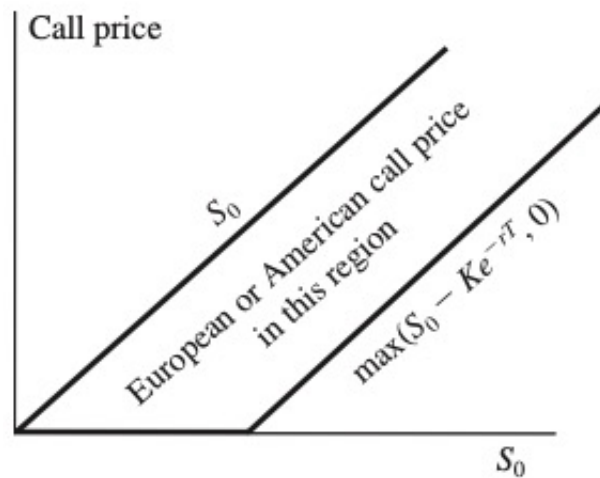
Because American call options are never exercised early when there are no dividends, they are equivalent to European call options, so that $C = c$. From equations (10.1) and (10.4), it follows that lower and upper bounds are given by

$$\max(S_0 - Ke^{-rT}, 0) \leq c \leq S_0$$

$$\max(S_0 - Ke^{-rT}, 0) \leq C \leq S_0$$

These bounds are illustrated in Figure 10.3.

No dividend, no reason to exercise early!



Source:
textbook, p.258

Figure 10.3 Bounds for European and American call options when there are no dividends

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ ($C = c$) (10.4)	?
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Problem 10.9

- *What is a lower bound for the price of a six-month call option on a non-dividend-paying stock when the stock price is \$80, the strike price is \$75, and the risk-free interest rate is 10% per annum?*

Answer 10.9

Setup

- Life of option (T) = 0.5
- Stock price today (S_0) = \$80
- Strike price (K) = \$75
- Risk-free rate (r) = 10%
- Find: lower bound for the call price (no dividend)

Answer 10.9

- Formula

$$c \geq \max(S_0 - Ke^{-rT}, 0) \quad (10.4)$$

- Then

$$LB = \text{Max}(S_0 - Ke^{-rT}, 0)$$

- Given

$$S_0 = 80, K = 75, r = 10\%, T = 0.5$$

- Therefore, the lower bound for the call option is

$$LB = \text{Max}(80 - 75e^{-0.10 \times 0.5}, 0) = \text{Max}(8.66, 0) = \$8.66$$

Lower bounds for puts on non-dividend-paying stocks

Portfolio C: one European put option plus one share $\rightarrow V_C(0) = p + S_0$

Portfolio D: an amount of cash equal to Ke^{-rT} (or equivalently a zero-coupon bond paying off K at time T) $\rightarrow V_D(0) = Ke^{-rT}$

If $S_T < K$, the option in portfolio C is exercised at option maturity and the portfolio becomes worth K . If $S_T > K$, the put option expires worthless and the portfolio is worth S_T at this time. Hence, portfolio C is worth

$$[\max(K - S_T, 0) + S_T] \rightarrow \max(S_T, K)$$

in time T . Portfolio D is worth K in time T . Hence, portfolio C is always worth as much as, and can sometimes be worth more than, portfolio D in time T . It follows that in the absence of arbitrage opportunities portfolio C must be worth at least as much as portfolio D today. Hence,

$$p + S_0 \geq Ke^{-rT}$$

or

$$p \geq Ke^{-rT} - S_0$$

Because the worst that can happen to a put option is that it expires worthless, its value cannot be negative. This means that

$$p \geq \max(Ke^{-rT} - S_0, 0) \quad (10.5)$$

Source:
textbook, p.253

From equations (10.3) and (10.5), lower and upper bounds for a European put option when there are no dividends are given by

$$\max(Ke^{-rT} - S_0, 0) \leq p \leq Ke^{-rT}$$

For an American put option on a non-dividend-paying stock, the condition

$$P \geq \max(K - S_0, 0)$$

must apply because the option can be exercised at any time. This is a stronger condition than the one for a European put option in equation (10.5). Using the result in equation (10.2), bounds for an American put option on a non-dividend-paying stock are

$$\max(K - S_0, 0) \leq P \leq K$$

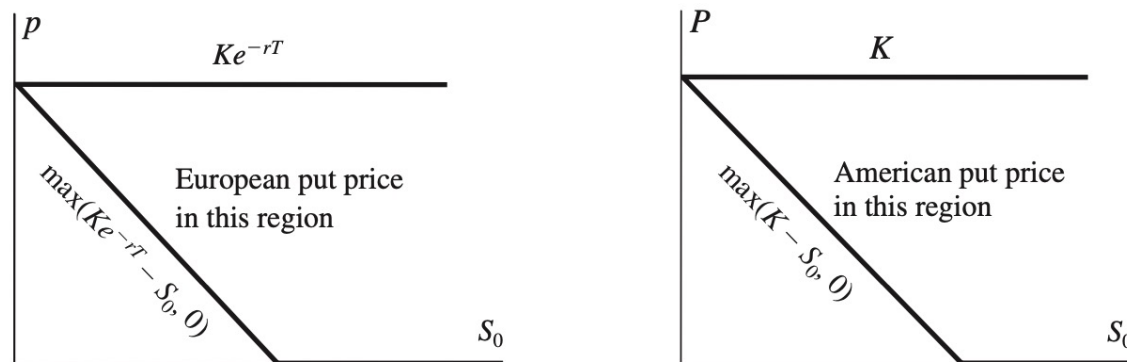


Figure 10.5 Bounds for European and American put options when there are no dividends

Source:
textbook, p.260

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$p \geq \max(Ke^{-rT} - S_0, 0)$ (10.5)
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$P \geq \max(K - S_0, 0)$
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Problem 10.12

- *A one-month European put option on a non-dividend-paying stock is currently selling for \$2.50. The stock price is \$47, the strike price is \$50, and the risk-free interest rate is 6% per annum. What opportunities are there for an arbitrageur?*

Answer 10.12

Setup

- Put option price (p) = \$2.50
- Life of option (T) = $\frac{1}{12}$
- Stock price today (S_0) = \$47
- Strike price (K) = \$50
- Risk-free rate (r) = 6%
- Find: lower bound for the put price (European/no dividend)
- Determine: whether the lower bound is violated
- Construct: the arbitrage strategy

Answer 10.12

$$p \geq \max(Ke^{-rT} - S_0, 0) \quad (10.5)$$

- The lower bound for the put option is

$$LB = \max(Ke^{-rT} - S_0, 0)$$

- In this case, $p = 2.50$, $T = \frac{1}{12}$, $S_0 = 47$, $K = 50$ and $r = 0.06$. The lower bound is

$$LB = \max(50e^{-0.06 \times 1/12} - 47, 0) = \max(2.75, 0) = \$2.75$$

- The mispricing is \$0.25 ($2.75 - 2.50$)
- Since $p < Ke^{-rT} - S_0$, then $p - Ke^{-rT} + S_0 < 0$
- Multiplying both sides by -1 gives

$$-p + Ke^{-rT} - S_0 > 0$$

VIOLATED!

Answer 10.12

• Thus the **arbitrage strategy** is

1. Buy the put option ($-p$)
2. Borrow the present value of K until time T (Ke^{-rT})
3. Buy the stock ($-S_0$)

Strategy	t=0	@T $S_T \geq 50$	@T $S_T < 50$
Buy put	-2.50	0	$50 - S_T$
Borrow Ke^{-rT}	+49.75	-50	-50
Buy stock	-47	$+S_T$	$+S_T$
Net CF	0.25	$-50 + S_T > 0$	0

Positive

Non-negative

Answer 10.12

- Regardless of what happens the arbitrageur will earn a profit of **\$0.25**. The net cash flow at time 0 is \$0.25 and the net cash flow at time T is non-negative.
- If the stock price is below \$50 in one month, the proceeds from the sale of the stock and the payoff on the put option will be sufficient to repay the loan. If the stock price is above \$50 in one month, the proceeds from the sale of the stock will be sufficient to repay the loan.

Recap: Put-call parity (without dividends)

Definition

- An important relationship between the prices of *European* put and call options that have the same strike price and time to maturity.

Formula

The components of portfolio A are worth c and Ke^{-rT} today, and the components of portfolio C are worth p and S_0 today. Hence,

$$c + Ke^{-rT} = p + S_0 \quad (10.6)$$

This relationship is known as *put-call parity*. It shows that the value of a European call with a certain exercise price and exercise date can be deduced from the value of a European put with the same exercise price and exercise date, and vice versa.

Source: textbook, p.254

Portfolio A: one European call option plus a zero-coupon bond that provides a payoff of K at time T $\rightarrow V_A(0) = c + Ke^{-rT}$

~~*Portfolio B:* one share of the stock~~

In portfolio A, the zero-coupon bond will be worth K at time T . If $S_T > K$, the call option is exercised at maturity and portfolio A is worth S_T . If $S_T < K$, the call option expires worthless and the portfolio is worth K . Hence, at time T , portfolio A is worth

$$\max(S_T, K)$$

Portfolio C: one European put option plus one share $\rightarrow V_C(0) = p + S_0$

~~*Portfolio D:* an amount of cash equal to Ke^{-rT} (or equivalently a zero-coupon bond paying off K at time T)~~

Identical results!

If $S_T < K$, the option in portfolio C is exercised at option maturity and the portfolio becomes worth K . If $S_T > K$, the put option expires worthless and the portfolio is worth S_T at this time. Hence, portfolio C is worth

$$\max(S_T, K)$$

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$p \geq \max(Ke^{-rT} - S_0, 0)$ (10.5)
Put-call parity. no dividend	$c + Ke^{-rT} = p + S_0$ (10.6)	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$P \geq \max(K - S_0, 0)$
Put-call parity. no dividend	?	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Recap: Put-call parity (without dividends)

- For European options, this is an **equality** because neither option can be exercised before maturity.
- For American options, things are more complicated because the put may be exercised early. ($C = c, P \geq p$)
- For American options (**inequality equations**)

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT} \quad (10.7)$$

Recap: Put-call parity (without dividends)

Proof

1. For the right-hand inequality: $C - P \leq S_0 - Ke^{-rT}$

- No dividends:

$$C = c, P \geq p$$

- Therefore:

$$C - P \leq c - p$$

- From European put-call parity:

$$c - p = Ke^{-rT} + S_0$$

- Therefore,

$$C - P \leq c - p = S_0 - Ke^{-rT}$$

Recap: Put-call parity (without dividends)

2. For the left-hand inequality: $S_0 - K \leq C - P$

Consider two portfolios:

- Portfolio A: one American call + K in cash ($C + K$)
- Portfolio B: one American put + one share of stock ($P + S_0$)

Scenario 1: If the put is exercised early at time t:

- *Portfolio B* = $S_t + (K - S_t) = K$
- While *Portfolio A* = $C_t + K \geq K$

Recap: Put-call parity (without dividends)

Scenario 2: If the put is not exercised early

- *Portfolio A* = $\max(S_T - K, 0) + K = \max(S_T, K)$
- *Portfolio B* = $\max(K - S_T, 0) + S_T = \max(S_T, K)$
- Thus, portfolio A is at least as much as Portfolio B at every relevant exercise date.
- Therefore, by no-arbitrage,

$$C + K \geq P + S_0$$

- So

$$S_0 - K \leq C - P$$

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$p \geq \max(Ke^{-rT} - S_0, 0)$ (10.5)
Put-call parity. no dividend	$c + Ke^{-rT} = p + S_0$ (10.6)	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$P \geq \max(K - S_0, 0)$
Put-call parity. no dividend	$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$ (10.7)	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Problem 10.16

- *The price of an American call on a non-dividend-paying stock is \$4. The stock price is \$31, the strike price is \$30, and the expiration date is in three months. The risk-free interest rate is 8%. Derive upper and lower bounds for the price of an American put on the same stock with the same strike price and expiration date.*

Answer 10.16

- **Call** option price (C) = \$4
- Life of option (T) = $\frac{3}{12}$
- Stock price today (S_0) = \$31
- Strike price (K) = \$30
- Risk-free rate (r) = 8%
- **Find: upper and lower bounds for the price of an **put** option (P)**

Answer 10.16

- From equation (10.7)

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT} \quad (10.7)$$

- In this case

$$31 - 30 \leq 4 - P \leq 31 - 30e^{-0.08 \times 0.25}$$

Or

$$1 \leq 4 - P \leq 1.59$$

Or

$$2.41 \leq P \leq 3.00$$

Lower and upper bounds are therefore \$2.41 and \$3.00.

Problem 10.17

- *Explain carefully the arbitrage opportunities in Problem 10.16 if the American put price is greater than the calculated upper bound.*

Answer 10.17

- The upper bound on the American put is given by

$$S_0 - K \leq C - P$$

- If (P is larger than its upper bound)

$$S_0 - K > C - P$$

- Then

$$C - P - S_0 + K < 0$$

- Multiplying both sides by -1 gives

$$-C + P + S_0 - K > 0$$

Answer 10.17

- Thus the arbitrage strategy is (based on the sign of the variables):
 (1) buy the call option; (2) sell the short option; (3) short sell the stock; (4) lend K

Strategy	t=0	@T $S_T \geq K$	@T $S_T < K$
Buy call	$-C$	$S_T - K$	0
Sell put	$+P$	0	$-(K - S_T)$
Short sell stock	$+S_0$	$-S_T$	$-S_T$
Lend K	$-K$	Ke^{rT}	Ke^{rT}
Net CF	> 0	$K(e^{rT} - 1) > 0$	$K(e^{rT} - 1) > 0$

Lower bounds for options on dividend-paying stocks

Portfolio A: one European call option plus an amount of cash equal to $D + Ke^{-rT}$

Portfolio B: one share

A similar argument to the one used to derive equation (10.4) shows that

$$P \geq \max\{[(S_0 - D) - Ke^{-rT}], 0\} \quad c \geq \max(S_0 - D - Ke^{-rT}, 0) \quad (10.8)$$

We can also redefine portfolios C and D as follows:

Portfolio C: one European put option plus one share

Portfolio D: an amount of cash equal to $D + Ke^{-rT}$

Replace S_0 with $(S_0 - D)$!

Owning the stock earns dividends;
owning the option does not.

A similar argument to the one used to derive equation (10.5) shows that

$$p \geq \max(D + Ke^{-rT} - S_0, 0) \quad (10.9)$$

$$p \geq \max\{[Ke^{-rt} - (S_0 - D)], 0\}$$

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$p \geq \max(Ke^{-rT} - S_0, 0)$ (10.5)
Put-call parity. no dividend	$c + Ke^{-rT} = p + S_0$ (10.6)	
Lower bound. with dividend	$c \geq \max(S_0 - D - Ke^{-rT}, 0)$ (10.8)	$p \geq \max(D + Ke^{-rT} - S_0, 0)$ (10.9)
Put-call parity. with dividend	?	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$P \geq \max(K - S_0, 0)$
Put-call parity. no dividend	$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$ (10.7)	
Lower bound. with dividend	?	?
Put-call parity. with dividend	?	

Problem 10.11

- *A four-month European call option on a dividend-paying stock is currently selling for \$5. The stock price is \$64, the strike price is \$60, and a dividend of \$0.80 is expected in one month. The risk-free interest rate is 12% per annum for all maturities. What opportunities are there for an arbitrageur?*

Answer 10.11

- Call option price (c) = \$5
- Life of option (T) = $\frac{4}{12}$
- Stock price today (S_0) = \$64
- Strike price (K) = \$60
- Risk-free rate (r) = 12%
- **Present value of dividends (D) = $\$0.8e^{-0.12 \times 1/12}$**
- **Find: the theoretical call bounds**
- **Determine: whether the bounds are violated**
- **Construct: the arbitrage strategy**

Answer 10.11

$$c \leq S_0 \quad \text{and} \quad C \leq S_0 \quad (10.1)$$

- The upper bound for the call option is $S_0 = 64$.
 - Conclusion: since the call price is traded at \$5, below the \$64 upper bound, there is **no violation**!
 - The lower bound for the call option on a dividend-paying stock is $LB = \text{Max}(S_0 - D - Ke^{-rT}, 0)$
 - Given $c = 5, T = \frac{4}{12}, S_0 = 64, K = 60, D = 0.79 (0.80e^{-0.12 \times \frac{1}{12}}), t = \frac{1}{12}$ and $r = 0.12$.

- The upper bound for the call option is $S_0 = 64$
- Conclusion: since the call price is traded at \$5, below the \$64 upper bound, there is **no violation**!
- The lower bound for the call option on a dividend-paying stock is $LB = \text{Max}(S_0 - D - Ke^{-rT}, 0)$

- Given

$$c = 5, T = \frac{4}{12}, S_0 = 64, K = 60, D = 0.79 \left(0.80e^{-0.12 \times \frac{1}{12}} \right), t = \frac{1}{12} \text{ and } r = 0.12.$$

Answer 10.11

- *The lower bound is*

$$LB = \text{Max} \left(64 - 0.80e^{-0.12 \times \frac{1}{12}} - 60e^{-0.12 \times \frac{4}{12}}, 0 \right) = \text{Max}(5.56, 0) \\ = \$5.56$$

- The present value of the strike price is \$57.65 and the present value of the dividend is \$0.79.
- With the call trading at \$5 the lower bound is **violated**. The mispricing is \$0.56 ($5.56 - 5$).

Answer 10.11

- Since

$$c < S_0 - D - Ke^{-rT}$$

- Then

$$c - S_0 + D + Ke^{-rT} < 0$$

- Multiplying both sides by -1 sides

$$-c + S_0 - D - Ke^{-rT} > 0$$

- **The arbitrage strategy** is (based on the sign of the variables)
- → (1) buy the call option; (2) sell short the stock; (3, 4) lend the present value of the dividend and strike price

Answer 10.11

Strategy	t=0	T=t	@T $S_T \geq 60$	@T $S_T < 60$
Buy call	-5.00	-0.8	$S_T - 60$	0
Short sell stock	+64.00		$-S_T$	$-S_T$
Lend Ke^{-rT}	-57.65		+60	+60
Lend D	-0.79	+0.8		
Net CF	0.56	0	0	$60 - S_T > 0$

- Regardless of what happens the arbitrageur will earn a profit of \$0.56. The net cash flow at time 0 is \$0.56 and the net cash flow at any other time is non-negative.

Answer 10.11

- The net cash flow in one month is zero since the \$0.79 invested for one month grows to \$0.80, which is enough to pay the dividend of \$0.80 due on the share that has been borrowed and sold.
- If the stock price is below \$60 in four months, the arbitrageur will have more than enough cash from the proceeds of the second loan to buy back the shorted stock.
- If the stock is above \$60 in four months, the payoff on the call option will compensate the arbitrageur for the additional cost incurred in buying back the shorted stock.

Recap: Put-call parity (with dividends)

- When dividends are expected, we can no longer assert that an American call option will not be exercised early. Sometimes it is optimal to exercise an American call immediately prior to an ex-dividend date.

Comparing the value at option maturity of the redefined portfolios A and C shows that, with dividends, the put–call parity result in equation (10.6) becomes

$$c + D + Ke^{-rT} = p + S_0 \quad (10.10)$$

Dividends cause equation (10.7) to be modified (see Problem 10.19) to

$$S_0 - D - K \leq C - P \leq S_0 - Ke^{-rT} \quad (10.11)$$

	European call c	European put p
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$p \geq \max(Ke^{-rT} - S_0, 0)$ (10.5)
Put-call parity. no dividend	$c + Ke^{-rT} = p + S_0$ (10.6)	
Lower bound. with dividend	$c \geq \max(S_0 - D - Ke^{-rT}, 0)$ (10.8)	$p \geq \max(D + Ke^{-rT} - S_0, 0)$ (10.9)
Put-call parity. with dividend	$c + D + Ke^{-rT} = p + S_0$ (10.10)	
	American call C	American put P
Upper bound (both)	$c \leq S_0$ and $C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$P \geq \max(K - S_0, 0)$
Put-call parity. no dividend	$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$ (10.7)	
Lower bound. with dividend	?	?
Put-call parity. with dividend	$S_0 - D - K \leq C - P \leq S_0 - Ke^{-rT}$ (10.11)	

Problem 10.14

- *The price of a European call that expires in six months and has a strike price of \$30 is \$2. The underlying stock price is \$29, and a dividend of \$0.50 is expected in two months and again in five months. Risk-free interest rates for all maturities are 10%. What is the price of a European put option that expires in six months and has a strike price of \$30?*

Answer 10.14

- Call option price (c) = \$2
- Life of option (T) = $\frac{6}{12}$
- Stock price today (S_0) = \$29
- Strike price (K) = \$30
- Risk-free rate (r) = 10%
- **Present value of dividends (D)** = $\$0.5(e^{-0.12 \times \frac{2}{12}} + e^{-0.12 \times \frac{5}{12}})$
- **Find: the fair put option price (p)**

Answer 10.14

- The put-call parity for a dividend-paying stock is

$$c + D + Ke^{-rT} = p + S_0 \quad (10.10)$$

- Rearranging the equation gives

$$p = c + D + Ke^{-rT} - S_0$$

- The present values of the strike price and the dividend are

$$Ke^{-rT} = 30e^{-0.1 \times 6/12} = \$28.54$$

$$D = 0.5e^{-0.1 \times 1/12} + 0.5e^{-0.1 \times 5/12} = 0.49 + 0.48 = \$0.97$$

- So the put price is

$$p = 2 + 28.54 + 0.97 - 29 = \$2.51$$

Problem 10.15

- *Explain carefully the arbitrage opportunities in Problem 10.14 if the European put price is \$3.*

Answer 10.15

- Call option price (c) = \$2
- Life of option (T) = $\frac{6}{12}$
- Stock price today (S_0) = \$29
- Strike price (K) = \$30
- Risk-free rate (r) = 10%
- **Present value** of dividends (D) = $\$0.5(e^{-0.12 \times \frac{2}{12}} + e^{-0.12 \times \frac{5}{12}})$
- **Find: the fair put option price ($p = 2.51$)**
- **Determine: whether the theoretical price is violated**
- **Construct: the arbitrage strategy**

Answer 10.15

$$c + D + Ke^{-rT} = p + S_0 \quad (10.10)$$

- If the put price is \$3.00, it **is too high (>\$2.51)** and an arbitrage profit of \$0.49 is available to the arbitrageur.

- If

$$c + D + Ke^{-rT} < p + S_0$$

- Then

$$p + S_0 - c - D - Ke^{-rT} > 0$$

- Thus the **arbitrage strategy** is (based on the sign on the variables): (1) sell the put option; (2) buy the call option; (3) lend the PV of D (D_1 till t_1 and t_2); (4) lend PV of Strike; (5) short sell the stock.

Answer 10.15

Strategy	t=0	T=t ₁	T=t ₂	@T S _T ≥ 30	@T S _T < 30
Sell put	+3.00			0	-(30 - S _T)
Buy call	-2.00			S _T - 30	0
Lend Ke^{-rT}	-28.54			+30	+30
Lend D ₁	-0.49	+0.50			
Lend D ₂	-0.48		+0.50		
Short sell stock	+29.00	-0.50	-0.50	-S _T	-S _T
Net CF	0.49	0	0	0	0

Answer 10.15

- Regardless of what happens the arbitrageur will earn a profit of \$0.49. The net cash flow at time 0 is \$0.49 and the net cash flow at time T is non-negative. At time t_1 , the liability for the first dividend of \$0.50 is met from the proceeds of the maturing loan made at time 0; similarly with the liability for the second dividend of \$0.50 at time t_2 .
- If the stock price is below \$30 in six months, the arbitrageur must pay out on the put option, the call option expires worthless and proceeds of \$30 are received from the maturing loan; the net cash flow is zero. If the stock price is above \$30 in six months, the call option generates a payoff for the arbitrageur, the put option expires worthless and proceeds of \$30 are received from the maturing loan; again, the net cash flow is zero.

	European call c	European put p
Upper bound (both)	$c \leq S_0 \quad \text{and} \quad C \leq S_0$ (10.1)	$p \leq Ke^{-rT}$ (10.3)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$p \geq \max(Ke^{-rT} - S_0, 0)$ (10.5)
Put-call parity. no dividend	$c + Ke^{-rT} = p + S_0$ (10.6)	
Lower bound. with dividend	$c \geq \max(S_0 - D - Ke^{-rT}, 0)$ (10.8)	$p \geq \max(D + Ke^{-rT} - S_0, 0)$ (10.9)
Put-call parity. with dividend	$c + D + Ke^{-rT} = p + S_0$ (10.10)	
	American call C	American put P
Upper bound (both)	$c \leq S_0 \quad \text{and} \quad C \leq S_0$ (10.1)	$P \leq K$ (10.2)
Lower bound. no dividend	$c \geq \max(S_0 - Ke^{-rT}, 0)$ (10.4)	$P \geq \max(K - S_0, 0)$
Put-call parity. no dividend	$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$ (10.7)	
Lower bound. with dividend	$C \geq \max(S_0 - D - Ke^{-rT}, S_0 - K, 0)$	$P \geq \max(D + Ke^{-rT} - S_0, K - S_0, 0)$
Put-call parity. with dividend	$S_0 - D - K \leq C - P \leq S_0 - Ke^{-rT}$ (10.11)	